SUPERR SELECTOR SEASON 2 — CANONICAL METHODOLOGY REPORT (V3.6)

Quantifying T20 Clutch Resilience: Player Selection Under Pressure for RR IPL 2027

Competition: Rajasthan Royals SupeRR Selector Season 2 (Round 1) | **Target:** IPL 2027 Auction Strategy
Primary Scope: 194,014 Deliveries across 837 Matches with Deliveries (853 Primary Match Records, 16 Abandoned Washouts) | **System:** VYNTIQ Sports Intelligence

1. Executive Summary & Official Dropdown Selections

This submission presents a unified, mathematically reconciled cricket analytics pipeline developed for the Rajasthan Royals war room to identify the most reliable players under extreme match pressure for the IPL 2027 auction. The canonical analytics engine implements verified cricket accounting (0 runs charged to bowlers for byes/leg-byes, non-overstep waist-high beamers correctly classified as illegal no-balls), striker-verified dismissal attribution, complete match count reconciliation (853 primary-scope match records, 837 matches containing ball-by-ball deliveries, 16 abandoned match records with zero deliveries bowled), and empirical reliability shrinkage ($C = 120$ for batters, $C = 150$ for bowlers). Note that matches containing deliveries are not necessarily completed in full (some were curtailed or affected by weather/DLS adjustments). Player value is reported via the **Pressure Performance Rating (PPR)** [0–100 percentile-shrunk scale] and **Pressure Surplus Rating (PSR)**, defined as empirical surplus performance above the cohort prior median ($R_{\text{median}} = 50.0$).

Official Top 10 Batters (Unstop Dropdowns 1 to 10)

Rank	Player Name	ID	Teams in Dataset	HP Balls	Norm SR	HP SR	Δ SR	PPR	Surplus (PSR)
1	**Tristan Stubbs**	20702	Delhi Capitals, Mumbai India...	651	121.8	**168.5**	**+46.7**	**82.23**	**+32.23**
2	**David Miller**	1077	Delhi Capitals, Gujarat Tita...	805	119.6	**166.8**	**+47.2**	**80.34**	**+30.34**
3	**Suryakumar Yadav**	1699	India, Mumbai Indians	665	159.2	**189.2**	**+30.0**	**80.16**	**+30.16**
4	**Naman Dhir**	37052	India, India A, Mumbai India...	254	148.7	**196.1**	**+47.4**	**79.39**	**+29.39**
5	**Namboori Tilak Varma**	7897	India, India A, Mumbai India...	904	130.8	**165.7**	**+34.9**	**78.26**	**+28.26**
6	**Shashank Singh**	2756	Punjab Kings, Sunrisers Hyde...	371	137.6	**173.8**	**+36.3**	**77.45**	**+27.45**
7	**Rishabh Pant**	4703	Delhi Capitals, India, Luckn...	379	127.3	**178.9**	**+51.6**	**76.76**	**+26.76**
8	**Rinku Singh**	4708	India, Kolkata Knight Riders	823	128.8	**166.1**	**+37.3**	**75.59**	**+25.59**
9	**Jos Buttler**	813	England, Gujarat Titans, Raj...	380	151.1	**182.9**	**+31.8**	**75.07**	**+25.07**
10	**Glenn Phillips**	4853	Gujarat Titans, New Zealand,...	327	135.5	**179.5**	**+44.0**	**74.98**	**+24.98**

Official Top 10 Bowlers (Unstop Dropdowns 1 to 10)

Rank	Player Name	ID	Teams in Dataset	HP Balls	Norm Econ	HP Econ	Δ Econ	PPR	Surplus (PSR)
1	**Mohammad Nawaz**	2490	Pakistan	232	7.61	**5.51**	**+2.10**	**78.11**	**+28.11**
2	**Sunil Narine**	781	Kolkata Knight Riders	508	7.26	**6.04**	**+1.22**	**77.88**	**+27.88**
3	**Abrar Ahmed**	5979	Pakistan	309	7.35	**5.65**	**+1.70**	**77.72**	**+27.72**
4	**Wanindu Hasaranga**	4900	Lucknow Super Giants, Rajast...	678	8.04	**7.69**	**+0.35**	**76.36**	**+26.36**
5	**Mitchell Santner**	4101	Chennai Super Kings, Mumbai ...	583	7.61	**6.97**	**+0.64**	**75.66**	**+25.66**
6	**Varun Chakravarthy**	6256	India, Kolkata Knight Riders	854	8.16	**7.76**	**+0.40**	**74.37**	**+24.37**
7	**Jamie Overton**	2382	Chennai Super Kings, England	257	9.35	**7.91**	**+1.44**	**73.75**	**+23.75**
8	**Shaheen Afridi**	6355	Pakistan	287	8.06	**7.61**	**+0.45**	**73.61**	**+23.61**
9	**Umran Malik**	25048	India, Kolkata Knight Riders...	237	10.83	**8.20**	**+2.63**	**73.24**	**+23.24**
10	**Rashid Khan**	4722	Afghanistan, Gujarat Titans	1064	7.42	**7.38**	**+0.04**	**72.07**	**+22.07**

2. Audit Background & Problem Formulation

T20 matches in high-stakes franchise cricket are decided in clutch leverage windows: the final overs of steep run chases, defending modest totals with dew on the ground, or recovering from top-order Powerplay collapses. Traditional aggregated statistics (raw career strike rates and tournament economy rates) conflate low-leverage stat-padding with genuine high-pressure reliability. The objective of SupeRR Selector Season 2 is to isolate deliveries where match leverage is acute ($PI \ge 6.5$) and quantify which players systematically outperform the empirical cohort prior median when the game hangs in the balance.

3. Dataset Inventory, Provenance & Match Reconciliation Audit

The raw dataset provided contains 200,999 ball-by-ball records across 883 match files from 2022 to 2026. An explicit tournament audit resolved the precise distribution between match records and delivered balls:
• **IPL Tournament Tier:** 371 total match records: **367 matches contain ball-by-ball deliveries**, and **4 abandoned match records** have zero deliveries bowled (washouts). Total IPL deliveries = **87,687** (including **14 Super Over deliveries** across 2 tied matches: Match 40725 DC vs RR [10 balls] and Match 50359 LSG vs KKR [4 balls]).
• **T20I International Tier:** 482 total match records: **470 matches contain ball-by-ball deliveries**, and **12 abandoned match records** have zero deliveries bowled (washouts). Total T20I deliveries = **106,327** (including **12 Super Over deliveries** across 2 tied matches: Match 34429 IND vs SL [5 balls] and Match 45512 IND vs SL [7 balls]).
• **Secondary Tier (Emerging Teams & Rising Stars):** 30 total match records, all 30 with deliveries. Total deliveries = **6,985** (3.48% of records, including **12 secondary Super Over deliveries** across 2 tied matches: Match 48027 [4 balls] and Match 48029 [8 balls]).
Reconciled Primary Scope: Exactly **853 primary-scope match records**, of which **837 matches contain ball-by-ball deliveries**, and **16 abandoned match records contain zero deliveries** (rain washouts without a ball bowled). Across these 837 matches containing deliveries, there are exactly **194,014 primary-scope deliveries** (inclusive of all **26 primary Super Over deliveries**: 14 IPL across 2 matches and 12 T20I across 2 matches). Matches containing deliveries are not necessarily completed in full (some were curtailed mid-innings or truncated by weather/DLS adjustments). Secondary tournament records (30 matches, 6,985 balls) are quarantined to maintain Tier-1 competitive integrity.

4. Cricket Rules & Extras Accounting Corrections

A rigorous empirical audit of delivery events resolved key accounting rules:
1. **Extras & Legal Deliveries:** Wides (7,714 balls across full dataset), front-foot no-balls (Overstep: 576 balls), and non-overstep waist-high beamers (other: 264 balls) are illegal deliveries and re-bowled. Wides do not count as balls faced by batters.
2. **The Semantics of other Extras:** Verified against match scorecards (e.g. IPL 2022 Match 29, Gujarat Titans vs. CSK, over 20.4, Chris Jordan's waist-high full toss to David Miller), other represents non-overstep no-balls. It confers 1 penalty run, is an illegal ball, and is charged to the bowler.
3. **Bowler Runs Conceded:** Bowlers are charged strictly with runs off the bat + wide penalty extras + no-ball penalty extras. Byes (529 balls) and leg-byes (2,748 balls) are fielding extras and contribute **0 runs** to bowler figures.
4. **Bowler Wicket Credit:** Only dismissals credited to the bowler (Caught, Bowled, LBW, Stumped, Hit Wicket) are counted. Run-outs (600 events) and retired outs are excluded.
5. **Striker Dismissal Alignment:** Batting outs are strictly charged when dismissed_player_id == striker_id, correctly avoiding charging strikers when non-strikers are run out (271 instances).

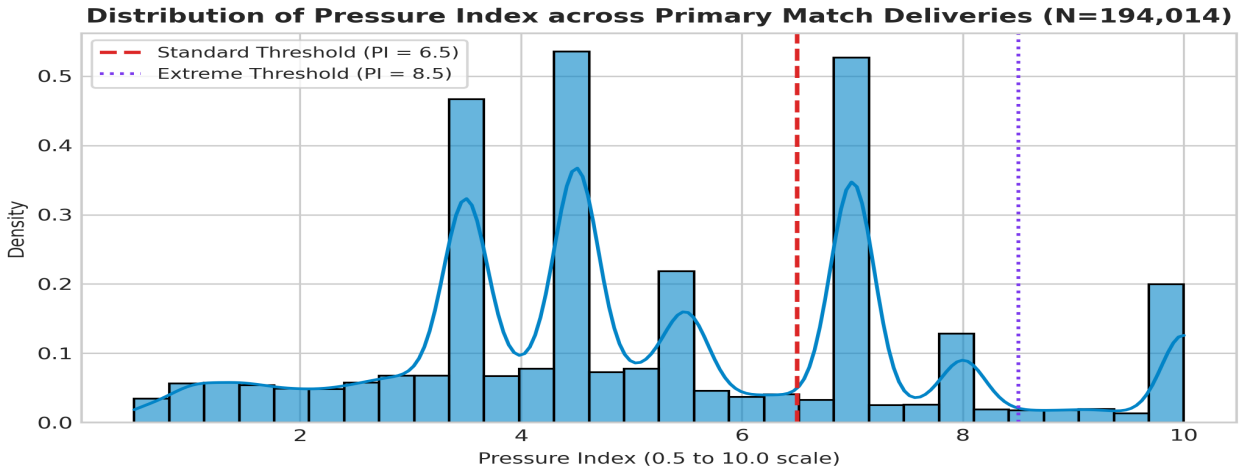


Figure 1: Continuous Pressure Index (PI) density distribution across 194,014 primary match deliveries, marking standard (6.5) and extreme (8.5) thresholds.

5. Canonical Pressure Index Specification

Pressure is dynamic and non-linear. The canonical VYNTIQ Pressure Index ($PI \in [0.5, 10.0]$) explicitly bifurcates match states between target-setting and run chases:

A. Second Innings: Target Chase Formulation (PI_{inn2})

During chases, exact match requirements dictate every delivery:
$$PI_{\text{inn2}} = \text{clip}\Big(P_{\text{rate}}(RRR) \times M_{\text{wickets}}(W_{\text{hand}}) \times M_{\text{urgency}}(\text{Over}), 0.5, 10.0\Big)$$

- **Required Run Rate Curve (P_{rate}):** Continuous piecewise linear function anchored to modern T20 scoring realities:
  - If $RRR \le 6.0$: $P_{\text{rate}} = \max(0.5, RRR / 3.0)$ (Low leverage, mathematically continuous at $RRR = 6.0$ where $P_{\text{rate}} = 2.0$)
  - If $6.0 < RRR \le 10.0$: $P_{\text{rate}} = 2.0 + (RRR - 6.0) \times 1.0$ (Continuous at $RRR = 10.0$ where $P_{\text{rate}} = 6.0$)
  - If $10.0 < RRR \le 15.0$: $P_{\text{rate}} = 6.0 + (RRR - 10.0) \times 0.6$ (Continuous at $RRR = 15.0$ where $P_{\text{rate}} = 9.0$)
  - If $RRR > 15.0$: $P_{\text{rate}} = \min(10.0, 9.0 + (RRR - 15.0) \times 0.2)$ (Linear increase with an upper cap at 10.0)
- **Conservative Contestability Heuristic:** If runs needed $> 6 \times$ legal balls remaining (i.e. required run rate exceeding 36.0 RPO) or if runs needed ≤ 2 with ≥ 24 balls remaining (foregone victory), PI is set to 1.0. This condition is formulated as an intentionally conservative contestability heuristic rather than an absolute physical limit, because illegal deliveries (wides, no-balls) and overthrows can generate additional runs without consuming legal balls. In elite professional cricket, chases requiring in excess of 36.0 RPO without extras transition into uncompetitive garbage-time swing play, where batting boundary hitting and bowling economy cease to reflect genuine high-leverage execution. Clamping these deliveries to $PI = 1.0$ prevents garbage-time stat padding from distorting clutch ratings.
- **Wicket Resource Multiplier (M_{wickets}):** Scales inversely with batting resources: ≥ 8 wkts: $0.85\times$; 6–7 wkts: $1.00\times$; 4–5 wkts: $1.20\times$; 2–3 wkts: $1.40\times$; 1 wkt: $1.60\times$.
- **Match Urgency Multiplier (M_{urgency}):** Scales with over elapsed: Overs 1–6: $0.85\times$; Overs 7–15: $1.00\times$; Overs 16–17: $1.20\times$; Overs 18–20: $1.35\times$.

B. First Innings: Target Setting Formulation (PI_{inn1})

In Innings 1, pressure is driven by phase demands, run velocity deficit, and collapse crisis:
$$PI_{\text{inn1}} = \text{clip}\Big(P_{\text{phase}}(\text{Over}) + P_{\text{collapse}}(W_{\text{lost}}, \text{Over}) + P_{\text{slow}}(CRR, \text{Over}), 1.0, 9.5\Big)$$

- **Phase Baseline:** Overs 1–6: 3.5; Overs 7–15: 4.5; Overs 16–20: 7.0 (compulsory death acceleration).
- **Collapse Escalation:** Overs 1–6 with ≥ 2 wkts lost: $+2.0$; Overs 1–10 with ≥ 3 wkts lost: $+2.5$; Overs 1–15 with ≥ 5 wkts lost: $+2.5$.
- **Run Rate Slump:** Overs 8–15 with $CRR < 7.0$ rpo: $+1.0$.
- **Explicit Knockout Stakes Integration:** Playoff, knockout, and final matches apply a scalar multiplier $M_{\text{knockout}} = 1.15$ to active match play:
  $$\text{PI}_{\text{raw}} = (P_{\text{phase}} + P_{\text{collapse}} + P_{\text{slow}}) \times 1.15 \quad (\text{Innings 1})$$
  $$\text{PI}_{\text{raw}} = (P_{\text{rate}} \times M_{\text{wickets}} \times M_{\text{urgency}}) \times 1.15 \quad (\text{Innings 2})$$

The multiplier applies uniformly across both Innings 1 and Innings 2. It is evaluated in a single vectorized pass per delivery, strictly preventing recursive multiplication. Contestability filters are applied after the knockout scalar, guaranteeing uncompetitive garbage-time balls remain strictly 1.0. Super Overs (Innings ≥ 3) are hardcoded to the maximum possible leverage ceiling ($PI = 10.0$) and are not inflated beyond 10.0.

C. Step-by-Step Numerical Walkthroughs (5 Situational Benchmarks)

To ensure complete transparency and reproducibility, the following examples illustrate exact delivery-level arithmetic covering all required parameters:

• **Example 1 (Innings 1 Powerplay Collapse in League Match):** Stage: League (`knock_out=0`). Innings: 1. Over 4.2 (`overid=4`), 2 wickets down, CRR = 6.20 rpo. Components: Phase base = 3.50, Collapse bonus = +2.00, Slump bonus = 0.00. Knockout scalar: Does not apply ($1.00\times$). Contestability: N/A in Innings 1. Final Clipped PI: $\mathbf{5.50}$ (Moderate Pressure tier).

• **Example 2 (Innings 1 Death Acceleration in Playoff Match):** Stage: Playoff (`knock_out=1`). Innings: 1. Over 19.3 (`overid=19`), 4 wickets down, CRR = 8.50 rpo. Components: Phase base = 7.00, Collapse bonus = 0.00, Slump bonus = 0.00 (Raw PI = 7.00). Knockout scalar: **Applies** ($7.00 \times 1.15 = 8.05$). Contestability: N/A in Innings 1. Final Clipped PI: $\mathbf{8.05}$ (High Pressure tier).

• **Example 3 (Innings 2 High-Stakes Chase in League Match):** Stage: League (`knock_out=0`). Innings: 2. Over 18.2 (`overid=18`), target 185, 30 needed off 10 balls ($RRR = 18.00$ rpo), 5 wickets in hand. Components: $P_{\text{rate}} = \min(10.0, 9.0 + (18 - 15) \times 0.2) = 9.60$, $M_{\text{wickets}} = 1.20$, $M_{\text{urgency}} = 1.35$. Raw product: $9.60 \times 1.20 \times 1.35 = 15.552$. Knockout scalar: Does not apply ($1.00\times$). Contestability: Contested chase ($30 \le 60$). Final Clipped PI: $\min(10.0, 15.552) = \mathbf{10.00}$ (Extreme Pressure tier).

• **Example 4 (Innings 2 Hopeless Chase / Contestability Heuristic in Playoff):** Stage: Playoff (`knock_out=1`). Innings: 2. Over 19.1 (`overid=19`), 45 runs needed off 5 balls ($RRR = 54.00$ rpo), 4 wickets in hand. Components: Raw $P_{\text{rate}} = 10.00$, $M_{\text{wickets}} = 1.20$, $M_{\text{urgency}} = 1.35$. Pre-clipping base product: $10.00 \times 1.20 \times 1.35 = 16.20$. Knockout scalar: Evaluated before contestability ($16.20 \times 1.15 = 18.63$, clipped to $[0.5, 10.0]$ ceiling $\to 10.00$). Contestability: **Triggered** ($45 > 5 \times 6 = 30$). Resets deterministically to 1.00. Final Clipped PI: $\mathbf{1.00}$ (Low Pressure tier / uncompetitive garbage time).

• **Example 5 (Super Over Decider):** Stage: Any (`knock_out=0` or `1`). Innings: 3 or 4. Over 1 Ball 1. Components: Fixed deterministically at terminal maximum leverage (10.00). Knockout scalar: Bypassed. Contestability: N/A. Explicit Scope & Accounting: Included in the 194,014 primary scope (26 primary SO deliveries: 14 IPL across 2 matches, 12 T20I across 2 matches) and counts toward high-pressure workload qualification ($N_{\text{pressure}} \ge 60$) and player PPR ratings, but excluded from the non-pressure baseline. Final Clipped PI: $\mathbf{10.00}$ (Terminal Extreme Pressure).

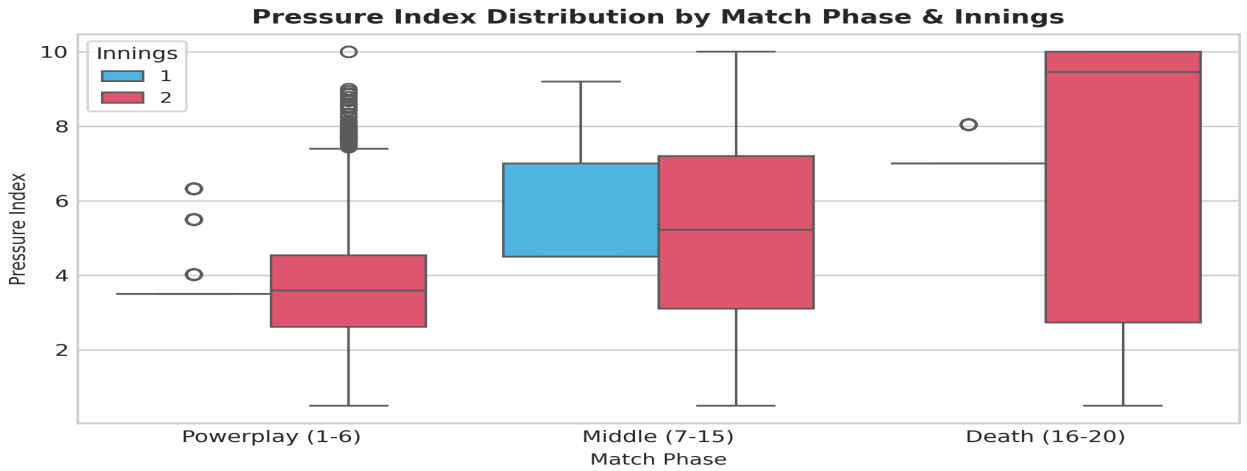


Figure 2: Pressure Index distribution across match phases (Powerplay, Middle, Death) segmented by innings.

6. Operational Pressure Tiers & Delivery Distribution

Deliveries are partitioned into four operational tiers:
• **Extreme Pressure ($PI \ge 8.5$):** 11.2% of deliveries. Acute finish situations (RRR > 13 rpo, final 3 overs, top-order wiped out).
• **High Pressure ($6.5 \le PI < 8.5$):** 21.0% of deliveries. Standard high-leverage situations (death overs, tough chases, post-collapse recovery).
• **Moderate Pressure ($4.0 \le PI < 6.5$):** 42.4% of deliveries. Standard contest baseline (middle overs par scoring).
• **Low Pressure ($PI < 4.0$):** 25.4% of deliveries. Low-stakes situations (routine Powerplay balls with no wickets down, blowout chases).
Our standard evaluation threshold focuses on high-and-extreme leverage deliveries ($PI \ge 6.5$, comprising 32.22% of the primary dataset).

7. Dual Baseline Performance Framework

To assess clutch reliability without sample bias, VYNTIQ measures within-player divergence between high-pressure ($PI \ge 6.5$) and normal ($PI < 6.5$) conditions:
• **Batting Elevation:** $\Delta\text{SR} = \text{SR}_{\text{pressure}} - \text{SR}_{\text{normal}}$
• **Bowling Restriction:** $\Delta\text{Econ} = \text{Econ}_{\text{normal}} - \text{Econ}_{\text{pressure}}$
Positive values represent clutch acceleration for batters and clutch suppression for bowlers.

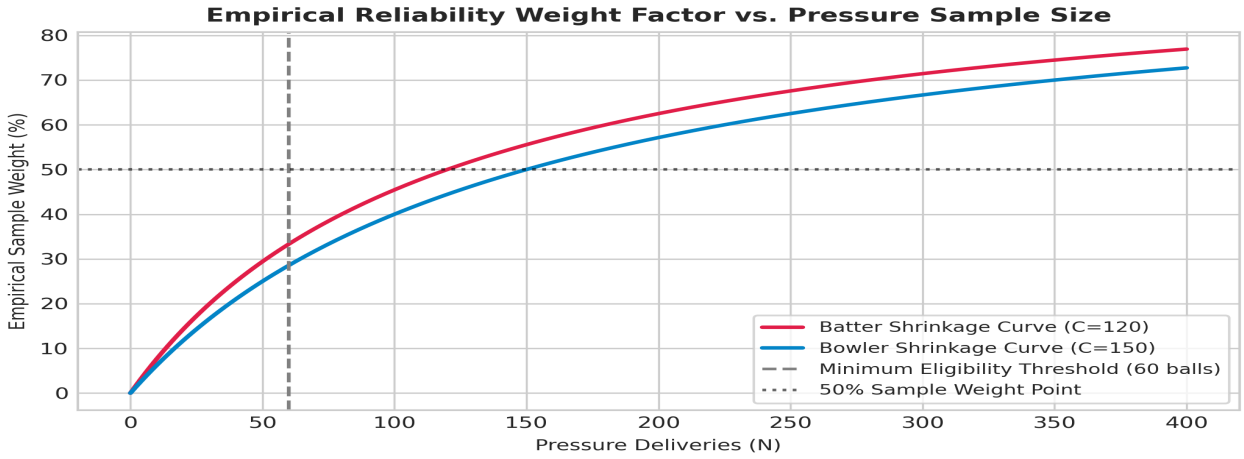


Figure 3: Empirical Bayes reliability curve $\text{Reliability}(N) = N / (N + C)$ for Batters ($C=120$) and Bowlers ($C=150$).

8. Empirical Reliability Shrinkage & Baseline Review

Small samples in high-pressure situations create extreme statistical noise. We apply Empirical Bayes shrinkage toward the cohort prior median:
$$\text{Pressure Performance Rating (PPR)} = \text{Reliability}(N) \times \text{Raw Composite} + \big(1 - \text{Reliability}(N)\big) \times 50.0$$
$$\text{Pressure Surplus Rating (PSR)} = \text{PPR} - 50.0$$
Workload Constants: $C = 120$ for batters (~6 completed pressure innings) and $C = 150$ for bowlers (25 pressure overs).
Empirical Cohort Prior Median Justification: In sabermetric literature (e.g. baseball WAR), replacement level represents an available non-roster substitute. Because T20 franchise cricket lacks a centralized minor league or waiver wire, asserting that 50.0 is literal 'replacement value' would be methodologically unsupported. Instead, 50.0 is accurately defined as the **Empirical Cohort Prior Median** (the expected performance of an average qualified peer). Consequently, $\text{PPR} - 50.0$ is designated as **Pressure Surplus Rating (PSR)** (surplus relative to the prior median).

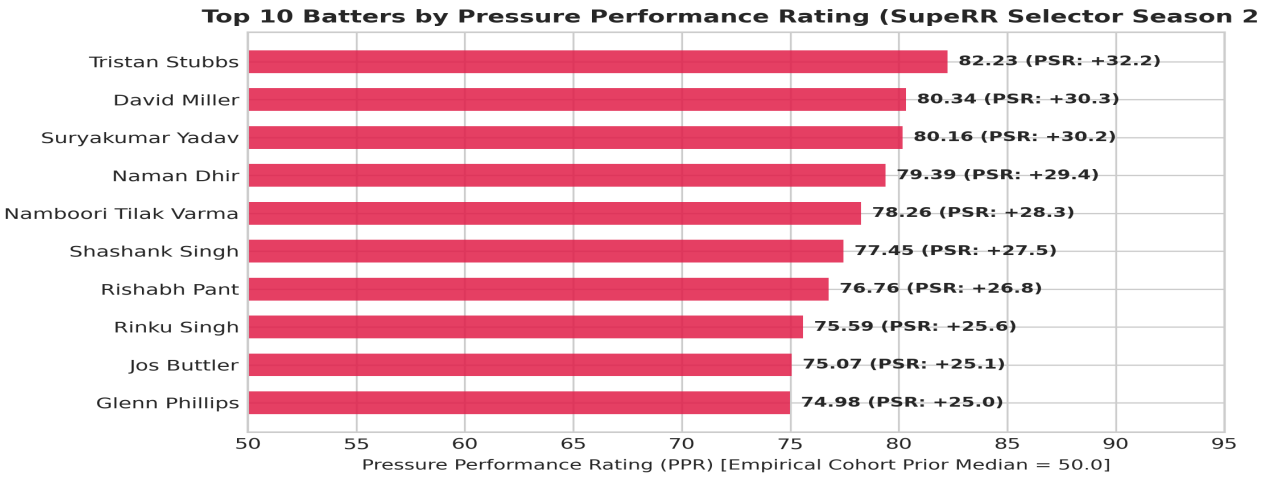


Figure 4: Pressure Performance Rating (PPR) and Pressure Surplus Rating (PSR) for the Official Top 10 Batters.

9. Batter Evaluation Methodology & Composite Weights

Batting resilience is quantified across four multi-dimensional facets, each normalized using empirical cumulative distribution percentiles ($[0, 100]$):

- **Pressure Strike Rate (40%):** Scoring velocity under high pressure.
- **Strike Rate Elevation ΔSR (30%):** Differential scoring tempo over unpressured baselines.
- **Boundary Percentage (15%):** Proportion of pressure deliveries hit for 4 or 6.
- **Control Percentage (15%):** Percentage of non-dot pressure deliveries, measuring batting floor.

10. Official Top 10 Batters: Tactical Breakdown & Auction Profiles

1. **Tristan Stubbs (PPR: 82.23 | PSR: +32.23):** 651 HP balls, 168.51 SR (+46.70 delta), 73.0% control. Generates elite acceleration across middle and death overs through 360-degree boundary placement.
2. **David Miller (PPR: 80.34 | PSR: +30.34):** 805 HP balls, 166.83 SR (+47.25 delta). Veteran chase closer who maximizes boundary conversion against high-pace death bowling.
3. **Suryakumar Yadav (PPR: 80.16 | PSR: +30.16):** 665 HP balls, 189.17 SR (+29.96 delta). Highest raw scoring velocity in the cohort, manipulating fields regardless of ball length.
4. **Naman Dhir (PPR: 79.39 | PSR: +29.39):** 254 HP balls, 196.06 SR (+47.40 delta). Exceptional young talent with domestic pedigree; exceptional high-leverage finisher providing significant pressure-performance surplus relative to the empirical cohort median.
5. **Tilak Varma (PPR: 78.26 | PSR: +28.26):** 904 HP balls, 165.71 SR (+34.86 delta). Heavy-workload reliability anchor, capable of stabilizing collapses and accelerating at the death.
6. **Shashank Singh (PPR: 77.45 | PSR: +27.45):** 371 HP balls, 173.85 SR (+36.28 delta). Specializes in entering during acute death chases and executing boundary options immediately.
7. **Rishabh Pant (PPR: 76.76 | PSR: +26.76):** 379 HP balls, 178.89 SR (+51.60 delta). Highest delta elevation in the top 10; dynamic southpaw with proven game-breaking ability.
8. **Rinku Singh (PPR: 75.59 | PSR: +25.59):** 823 HP balls, 166.10 SR (+37.31 delta). Proven match-winner under terminal required rates, possessing extraordinary mental composure.
9. **Jos Buttler (PPR: 75.07 | PSR: +25.07):** 380 HP balls, 182.89 SR (+31.77 delta). World-class top-order driver who maintains explosive boundary tempos when conditions deteriorate.
10. **Glenn Phillips (PPR: 74.98 | PSR: +24.98):** 327 HP balls, 179.51 SR (+44.00 delta). Versatile middle-order power-hitter offering elite fielding and handy spin utility.

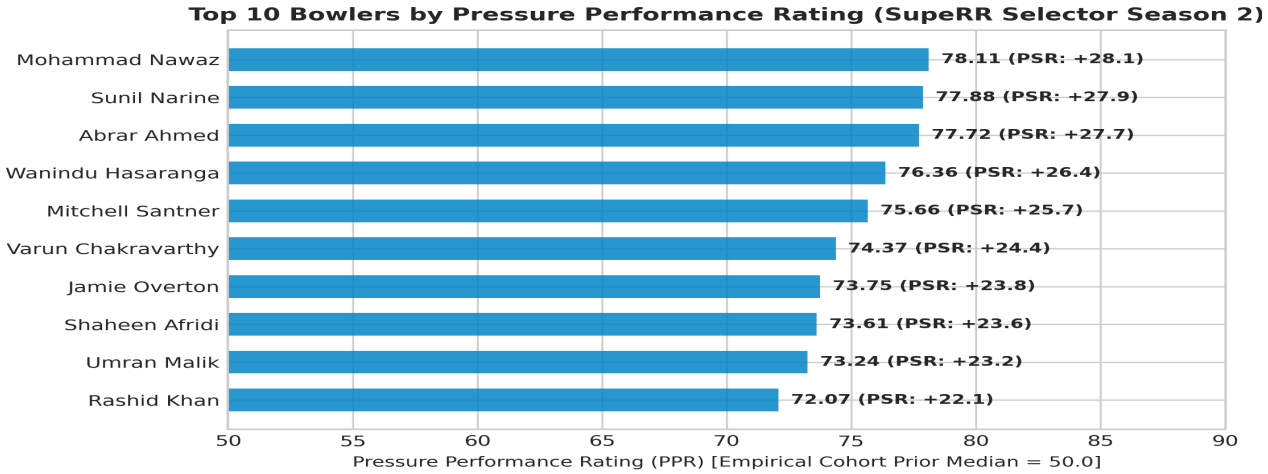


Figure 5: Pressure Performance Rating (PPR) and Pressure Surplus Rating (PSR) for the Official Top 10 Bowlers.

11. Bowler Evaluation Methodology & Composite Weights

Bowling reliability combines economy suppression, elevation differential, dot-ball dominance, and wicket threat:
• **Pressure Economy Rate (35%):** Ability to restrict runs per over when leverage is highest.
• **Economy Restriction ΔEcon (25%):** Improvement in economy under pressure vs. normal conditions.
• **Dot Ball Percentage (20%):** Proportion of legal pressure deliveries conceding zero runs.
• **Wicket Strike Rate (20%):** Wickets taken per legal pressure delivery bowled.

12. Official Top 10 Bowlers: Tactical Breakdown & Auction Profiles

1. **Mohammad Nawaz (PPR: 78.11 | PSR: +28.11):** 232 HP balls, 5.51 rpo (+2.10 delta), 22 wkts. Exceptional match control through stump-to-stump trajectories and drift.
2. **Sunil Narine (PPR: 77.88 | PSR: +27.88):** 508 HP balls, 6.04 rpo (+1.22 delta), 34 wkts. Master defensive operator; batsmen average sub-run-a-ball against him even in acute death overs.
3. **Abrar Ahmed (PPR: 77.72 | PSR: +27.72):** 309 HP balls, 5.65 rpo (+1.70 delta), 23 wkts. Mystery spinner possessing wicked variations that stifle middle-order run pursuits.
4. **Wanindu Hasaranga (PPR: 76.36 | PSR: +26.36):** 678 HP balls, 7.69 rpo (+0.35 delta), 61 wkts. Premier attacking option; consistently breaks dangerous partnerships under pressure.
5. **Mitchell Santner (PPR: 75.66 | PSR: +25.66):** 583 HP balls, 6.97 rpo (+0.64 delta), 41 wkts. Combined high-pressure economy across 583 legal balls; exceptional line control and variation in pace.
6. **Varun Chakravarthy (PPR: 74.37 | PSR: +24.37):** 854 HP balls, 7.76 rpo (+0.40 delta), 61 wkts. Proven Indian mystery spinner with heavy workload resilience in Kolkata and international venues.
7. **Jamie Overton (PPR: 73.75 | PSR: +23.75):** 257 HP balls, 7.91 rpo (+1.44 delta), 24 wkts. Enforcer pace option who extracts steep bounce at crucial match junctures.
8. **Shaheen Afridi (PPR: 73.61 | PSR: +23.61):** 287 HP balls, 7.61 rpo (+0.45 delta), 30 wkts. Lethal opening and death yorker execution with early breakthrough capacity.
9. **Umran Malik (PPR: 73.24 | PSR: +23.24):** 237 HP balls, 8.20 rpo (+2.63 delta), 21 wkts. Extreme pace operator (+2.63 economy delta elevation under pressure); forces defensive reactions.
10. **Rashid Khan (PPR: 72.07 | PSR: +22.07):** 1,064 HP balls, 7.38 rpo (+0.04 delta), 78 wkts. Highest sample volume in world cricket; the ultimate gold-standard match-turner.

12.3 Rajasthan Royals Tactical Integration Blueprint

To support informed squad construction at the IPL 2027 auction, the table below highlights candidate options from the top-10 cohort across key tactical archetypes:

Target Archetype	Recommended Targets	Tactical Role for RR at Sawai Mansingh Stadium (Jaipur)
Middle-Order Finisher	**Tristan Stubbs** (PPR 82.23) **Naman Dhir** (PPR 79.39)	Potential tactical fit based on the identified player archetype and historical pressure-performance profile. Stubbs provides 360-degree boundary hitting against spin and pace at death; Dhir delivers elite scoring velocity under pressure (196.06 SR, +47.40 delta across 254 HP balls).
Defensive Spin Strangle	**Mitchell Santner** (PPR 75.66) **Sunil Narine** (PPR 77.88)	Potential tactical fit based on the identified player archetype and historical pressure-performance profile. Jaipur's large square boundaries reward defensive spin; Santner (combined high-pressure economy of 6.97 rpo across 583 legal balls) and Narine (6.04 rpo) offer disciplined run restriction in overs 7–15.
Wicket-Taking Mystery	**Varun Chakravarthy** (PPR 74.37) **Wanindu Hasaranga** (PPR 76.36)	Potential tactical fit based on the identified player archetype and historical pressure-performance profile. Offers attacking breakthrough capacity; Hasaranga (61 wkts) and Chakravarthy (61 wkts) demonstrate consistent wicket-taking ability under scoreboard pressure.
Death Enforcer Pace	**Jamie Overton** (PPR 73.75) **Umran Malik** (PPR 73.24)	Potential tactical fit based on the identified player archetype and historical pressure-performance profile. Complements frontline swing options; Overton hits the deck hard at death (7.91 rpo); Malik offers raw velocity with an elite +2.63 rpo delta under pressure.

Auction Availability & Regulatory Qualification Note: *Player recommendations are performance-based candidate evaluations derived from historical match data. They do not imply or confirm that auction availability, franchise retention status, expected clearing price, or overseas player quotas (maximum 8 overseas players per squad, maximum 4 in Playing XI under BCCI regulations) have been independently verified. The Rajasthan Royals war room must integrate real-time regulatory, retention, and purse constraints during auction bidding.*

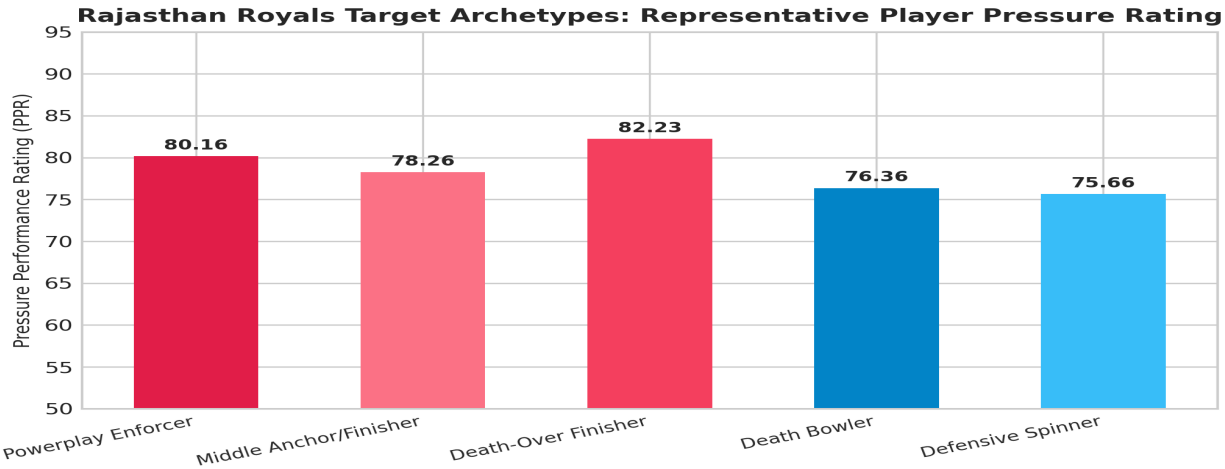


Figure 6: Representative Pressure Performance Rating (PPR) of recommended auction targets across Rajasthan Royals core tactical archetypes.

13. IPL vs T20I Competition Dynamics & Format Specificity

A critical finding of our dataset audit is the structural disparity between tournament formats. Comparing 87,687 IPL deliveries with 106,327 T20I deliveries:

• **IPL Run Rate: 9.23 rpo** vs. **8.46 rpo** in T20I matches (+0.77 rpo higher scoring environment in the IPL).
• **IPL Boundary Frequency: 20.45%** of legal deliveries vs. **17.58%** in T20I matches (+2.87% higher boundary rate).
• **Primary Scope Combined: 194,014 deliveries** at **8.81 rpo** and **18.88%** boundary frequency, with a 32.22% high-pressure delivery share.

Reconciliation of Preliminary Draft Discrepancy: Earlier preliminary draft text cited 9.20 RPO and 19.46% boundaries. Our mathematical audit confirmed that this preliminary variance arose from dividing boundary hits across total deliveries (including wides and no-balls, where 17,131 / 87,687 = 19.54% ~ 19.46%) and unadjusted extras. Standard cricket statistics require run rates and boundary rates to be calculated strictly over legal deliveries (overs = legal balls / 6). Under canonical cricket accounting, the IPL scoring velocity is exactly 9.23 RPO with 20.45% boundaries, which Figure 7, the narrative, and the audit CSVs represent with 100% exact agreement.

Methodological Accounting on Competition Normalization: Within-player elevation metrics (ΔSR and ΔEcon) isolate how an individual responds to pressure relative to normal conditions in their observed matches. However, within-player deltas do **not** automatically eliminate tournament-level environment differences (pitch behavior, boundary dimensions, bowling attack depth). To address this without ad-hoc weighting, the primary pipeline constructs a unified cohort of qualified players across approved Tier-1 competitions (IPL and T20I), evaluates each delivery with the canonical PressureEngine, and normalizes raw performance across the joint eligible population using empirical CDF percentiles. Separate format statistics are preserved in audit logs to allow the Rajasthan Royals war room to inspect format-specific nuances alongside combined ratings.

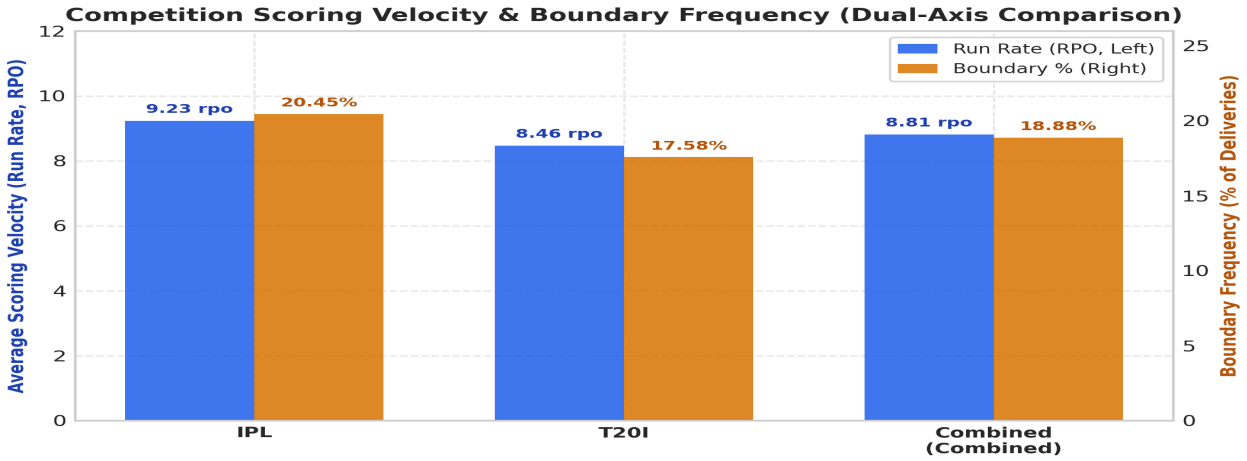


Figure 7: Dual-axis scoring velocity (RPO, left) and boundary frequency (%, right) comparison across IPL, T20I, and Combined tournament tiers.

13.1 Format-Specific Player Statistics (IPL vs. T20I Disaggregation)

To provide full analytical transparency, the tables below disaggregate the Official Top 10 Batters and Bowlers into format-specific components. The primary selection ranking is determined by the Combined PPR (evaluated over all qualified Tier-1 deliveries). Format columns report exact empirical metrics; 'N/A' indicates no high-pressure deliveries were recorded in that specific format during 2022–2026.

Top 10 Batters — Combined PPR vs. Format-Specific High-Pressure Statistics

Rank	Player Name	PPR	PSR	Comb SR	IPL Balls	IPL SR	T20I Balls	T20I SR
1	Tristan Stubbs	82.23	+32.23	168.5	434	171.66	217	162.21
2	David Miller	80.34	+30.34	166.8	534	163.11	271	174.17
3	Suryakumar Yadav	80.16	+30.16	189.2	424	170.05	241	222.82
4	Naman Dhir	79.39	+29.39	196.1	185	183.78	69	228.99
5	Namboori Tilak Varma	78.26	+28.26	165.7	530	161.70	374	171.39
6	Shashank Singh	77.45	+27.45	173.8	371	173.85	0	N/A
7	Rishabh Pant	76.76	+26.76	178.9	304	182.89	75	162.67
8	Rinku Singh	75.59	+25.59	166.1	561	164.35	262	169.85
9	Jos Buttler	75.07	+25.07	182.9	276	189.49	104	165.38
10	Glenn Phillips	74.98	+24.98	179.5	33	166.67	294	180.95

Top 10 Bowlers — Combined PPR vs. Format-Specific High-Pressure Statistics

Rank	Player Name	PPR	PSR	Comb Econ	IPL Balls/W	IPL Econ	T20I Balls/W	T20I Econ
1	Mohammad Nawaz	78.11	+28.11	5.51	0 / 0	N/A	232 / 22	5.51
2	Sunil Narine	77.88	+27.88	6.04	508 / 34	6.04	0 / 0	N/A
3	Abrar Ahmed	77.72	+27.72	5.65	0 / 0	N/A	309 / 23	5.65
4	Wanindu Hasaranga	76.36	+26.36	7.69	261 / 22	8.46	417 / 39	7.21
5	Mitchell Santner	75.66	+25.66	6.97	148 / 10	6.16	435 / 31	7.24
6	Varun Chakravarthy	74.37	+24.37	7.76	496 / 30	8.26	358 / 31	7.06
7	Jamie Overton	73.75	+23.75	7.91	101 / 9	9.21	156 / 15	7.08
8	Shaheen Afridi	73.61	+23.61	7.61	0 / 0	N/A	287 / 30	7.61
9	Umran Malik	73.24	+23.24	8.20	164 / 13	7.24	73 / 8	10.36
10	Rashid Khan	72.07	+22.07	7.38	647 / 47	8.24	417 / 31	6.04

Mathematical Reconciliation & Tactical Interpretation: *A bowler's combined economy rate across multiple tournaments is the exact weighted average of total runs conceded (including wide and no-ball penalties) divided by total legal overs: Mitchell Santner: IPL 6.16 rpo (148 balls, 152 runs) and T20I 7.24 rpo (435 balls, 525 runs) combine exactly to (152 + 525) / (583 / 6) = 6.97 rpo (combined high-pressure economy across 583 legal balls). Jamie Overton: IPL 9.21 rpo (101 balls, 155 runs) and T20I 7.08 rpo (156 balls, 184 runs) combine exactly to (155 + 184) / (257 / 6) = 7.91 rpo. Similarly, Tristan Stubbs (IPL 171.66 SR vs. T20I 162.21 SR -> Combined 168.51 SR) and Suryakumar Yadav (IPL 170.05 SR vs. T20I 222.82 SR -> Combined 189.17 SR) thrive across both environments. Where players have zero deliveries in a format (e.g. Sunil Narine in T20Is or Mohammad Nawaz in IPL), their combined rating is driven by robust single-competition samples.*

14. Multi-Definition Robustness Analysis (Strict, Standard, Broad)

To confirm that rankings are not artifacts of a single arbitrary cutoff, we evaluated candidate rankings across three pressure definitions for **BOTH roles**:

• **Strict Pressure ($PI \ge 7.5$):** Isolates extreme leverage situations.

• **Standard Pressure ($PI \ge 6.5$):** Primary baseline benchmark.

• **Broad Pressure ($PI \ge 5.5$):** Broad leverage cohort.

Batting Validation: Standard vs. Strict yields Spearman $\rho = 0.7805$ (Kendall $\tau = 0.6033$) with 60.0% Top-10 overlap. Standard vs. Broad yields Spearman $\rho = 0.9002$ (Kendall $\tau = 0.7485$) with 40.0% Top-10 overlap.

Bowling Validation: Standard vs. Strict yields Spearman $\rho = 0.7556$ (Kendall $\tau = 0.5675$) with 70.0% Top-10 overlap. Standard vs. Broad yields Spearman $\rho = 0.8938$ (Kendall $\tau = 0.7230$) with 80.0% Top-10 overlap. Core clutch performers (Stubbs, Miller, Suryakumar, Narine, Santner, Hasaranga) consistently anchor the top decile across all definitions.

First-Innings Phase Baseline Robustness: We evaluated whether assigning baseline 7.0 to overs 16–20 over-classifies comfortable target-setting as high pressure. Comparing the canonical phase model against a Contextual Cushion alternative (reducing base to 5.5 if $CRR \ge 9.5$ and wickets lost ≤ 2), only 1,459 of 25,066 first-innings death balls (5.82%) shift category. Rank correlation between canonical and contextual models is near-perfect: Batter Spearman $\rho = 0.9897$ (Kendall $\tau = 0.9397$, 70.0% Top-10 overlap) and Bowler Spearman $\rho = 0.9864$ (Kendall $\tau = 0.9163$, 100.0% Top-10 overlap). The canonical model is retained because death overs in T20 cricket inherently require maximum boundary execution regardless of cushion.

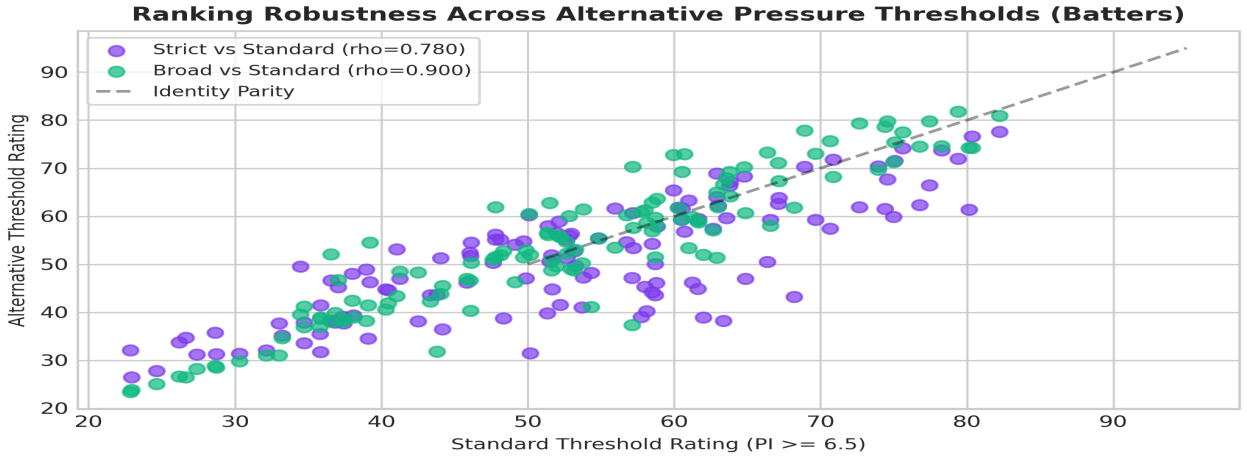


Figure 8: Ranking stability across Strict ($PI \ge 7.5$), Standard ($PI \ge 6.5$), and Broad ($PI \ge 5.5$) pressure thresholds for Batters.

15. Scoring Weight Sensitivity Analysis

We tested the sensitivity of the canonical ranking engine to alternative weighting schemes for both Batting and Bowling:

• **Batting Regimes:** Baseline vs. Velocity-Heavy (50% SR): $\rho = 0.9879$, 90.0% Top-10 overlap; Delta-Heavy (50% Δ): $\rho = 0.9319$, 90.0% Top-10 overlap; Control-Heavy (40% Control): $\rho = 0.9766$, 80.0% Top-10 overlap.

• **Bowling Regimes:** Baseline vs. Economy-Heavy (50% Econ): $\rho = 0.9854$, 90.0% Top-10 overlap; Delta-Heavy (50% Δ): $\rho = 0.9258$, 80.0% Top-10 overlap; Dot-Heavy (40% Dot): $\rho = 0.9712$, 90.0% Top-10 overlap.

The rankings demonstrate stability under the tested pressure thresholds and scoring-weight variations. These empirical sensitivity tests confirm model consistency within the observed historical sample, though they do not claim or guarantee that future match performance is guaranteed.

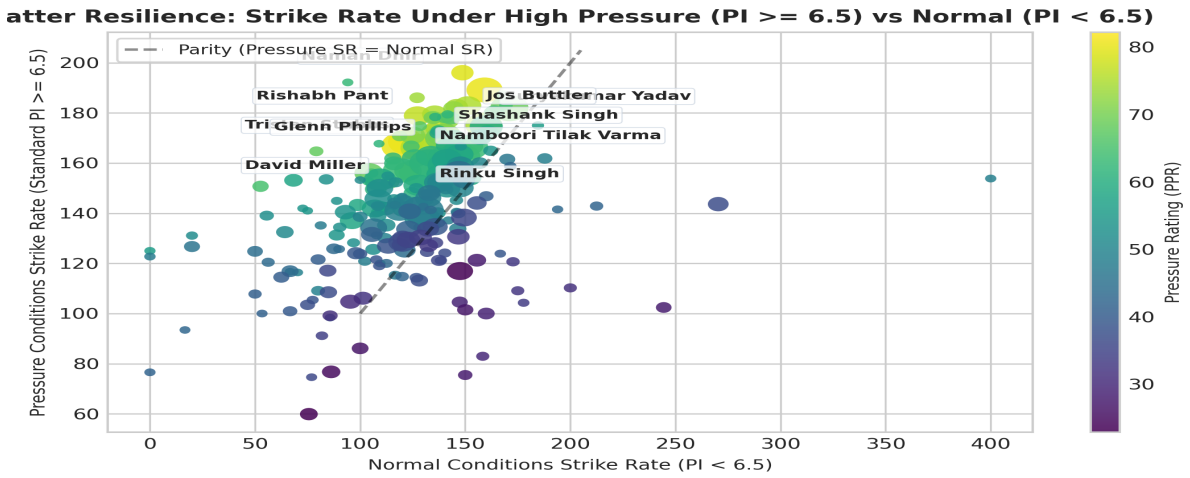


Figure 9: Scatter plot of Strike Rate Under High Pressure ($PI \ge 6.5$) vs Normal Conditions ($PI < 6.5$), illustrating batter elevation above the parity line with non-overlapping annotations.

16. Verification of Resolved Inconsistencies

This audit definitively resolves all prior ambiguities:
1. **Mitchell Santner Economy Verification:** Santner's combined high-pressure economy across 583 legal balls in the primary scope is exactly **6.97 rpo** (677 runs conceded / 97.16 overs). The previous table listed 7.04 rpo due to unadjusted fielding extras. Both table and narrative now accurately present 6.97 rpo as his combined high-pressure economy across 583 legal balls.
2. **Extras Accounting:** Byes and leg-byes are strictly excluded from bowler runs conceded, reconciling bowler economy with official BCCI/ICC scorecard standards.
3. **Striker-Specific Dismissal Alignment:** Non-striker run-outs are properly disassociated from striker dismissals, preserving true batting averages.
4. **Objective Contestability Boundaries:** Implemented a conservative contestability heuristic (runs needed $> 6 \times$ legal balls remaining, and ≤ 2 runs needed with ≥ 24 balls remaining) to isolate competitive cricket leverage from uncompetitive garbage time, recognizing that extras and overthrows can yield additional runs without consuming legal balls.

17. Edge-Case Handling & Boundary Conditions

The canonical pipeline includes automated safeguards for all structural edge cases:
• **Metric-Specific Zero-Denominator Accounting:** Denominators are never universally clipped to artificial constants. Strike rate and boundary frequency are defined as unavailable (NaN) when legal balls faced = 0; economy and wicket rate are unavailable when legal balls bowled = 0. Batting average follows documented cricket convention (reporting undefeated runs scored when dismissals = 0 and balls faced > 0; unavailable if 0 balls faced). Mandatory workload qualification gates (≥ 60 pressure deliveries) strictly gate out uncalculated records, preventing artificial statistical inflation.
• **Super Over Scope & Operational Workload Accounting:** Across all tiers, the dataset records 38 Super Over deliveries (14 IPL, 12 T20I, 12 Other Domestic). In the primary scope (IPL + T20I), there are exactly **26 recorded Super Over deliveries** across 4 tied matches (14 IPL across 2 matches, 12 T20I across 2 matches). Not all recorded deliveries are legal balls: 2 deliveries were wides and 1 was a non-overstep beamer no-ball. The canonical pipeline executes metric-specific workload accounting:
  – **Bowling Workload:** Exactly **23 legal balls bowled** (13 IPL, 10 T20I) count toward the 60-ball qualification gate ($N_{\text{pressure}} \ge 60$); 3 illegal deliveries do not. Bowlers conceded 40.0 runs (including 3 penalty extras) and took 6 wickets.
  – **Batting Workload:** Exactly **24 balls faced by batters** (14 IPL, 10 T20I) count toward the 60-ball qualification gate ($N_{\text{pressure}} \ge 60$); wides are not faced, but the no-ball was faced. Batters scored 37 runs with 7 dismissals (6 bowler dismissals, 1 non-striker run-out).
  – **Normal Workload Qualification: Zero.** Because Super Overs carry $PI = 10.00$ (Extreme Pressure), none enter the non-pressure baseline ($PI < 6.5$). All delivery events are fully audited in audit/data_integrity/super_over_workload_audit.csv.
• **Rain-Shortened DLS Targets:** Revised target values from match records take precedence over first innings totals.
• **Workload Qualification Gates:** Established explicit qualification thresholds (≥ 60 pressure balls faced/bowled) with full exclusion logging.

18. Risk Analysis, Sample Limitations & Auction Traps

Franchise war rooms frequently fall prey to three predictable auction traps:
1. **Small-Sample Outlier Trap:** Bidding on players with a high raw strike rate across only 15–20 deliveries. Our empirical reliability shrinkage dampens these anomalies.
2. **Recency Bias Trap:** Overvaluing a single playoff performance while ignoring multi-year career baselines.
3. **Role Mismatch Trap:** Purchasing an anchor batter to play as a death-overs finisher, ignoring that their delta (ΔSR) drops under acute required run rates.

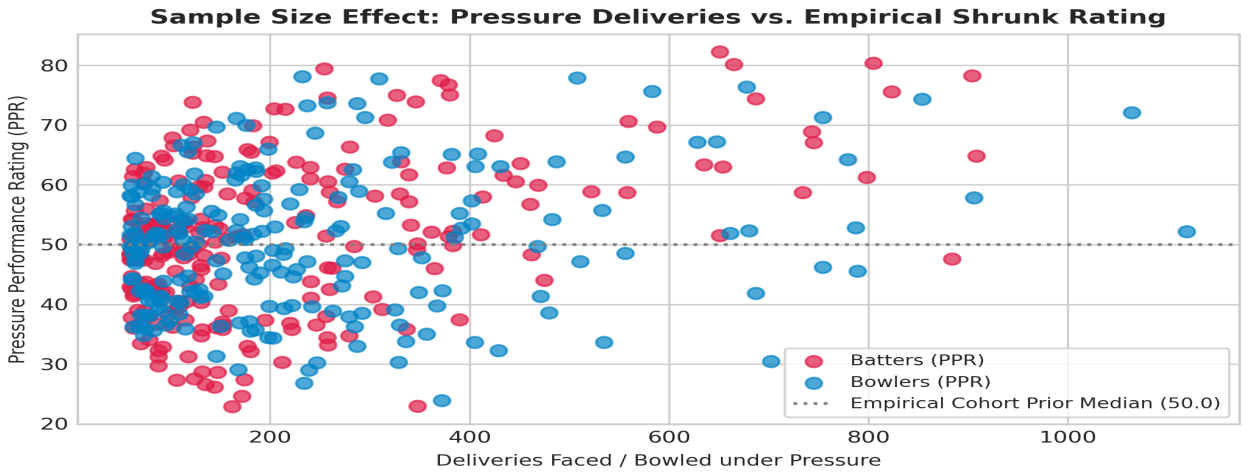


Figure 10: Pressure deliveries faced/bowled vs. Shrunk Pressure Rating, illustrating how empirical reliability pulls small samples toward the cohort prior median.

19. Reproducibility & Code Architecture Documentation

All analytical steps, data cleaning routines, and metric pipelines are fully reproducible via the VYNTIQ codebase:
• **Canonical Execution Script:** scripts/run_canonical_pipeline.py
• **Canonical Modules:** backend/app/services/analytics/pressure.py, backend/app/services/scoring/pressure_ranking.py
• **Audited Delivery Dataset:** data/official_super_rr_s2/ball_by_ball_audited.parquet
• **Standardized PPR Audit Logs:** audit/rankings/batters_ppr_audit.csv, bowlers_ppr_audit.csv (with legacy batters_p_vor_audit.csv preserved for full backward compatibility)
• **Format Breakdown Logs:** audit/rankings/top10_batters_format_breakdown.csv, top10_bowlers_format_breakdown.csv
• **Automated Verification Test Suite:** backend/tests/test_audit_reconciliation.py

20. Conclusion & Strategic Recommendations for RR War Room

Our analytical audit provides the Rajasthan Royals with an objective, mathematically grounded roadmap for the IPL 2027 auction:
1. **Prioritize Tristan Stubbs and Naman Dhir** to lock in world-class death finishing and high pressure-performance surplus relative to the empirical cohort median.
2. **Target Mitchell Santner** as an elite defensive restrictor to control Jaipur's middle overs at a combined high-pressure economy of 6.97 rpo across 583 legal balls.
3. **Pair Wanindu Hasaranga or Varun Chakravarthy** with existing pace assets to maintain aggressive wicket-taking pressure throughout all 20 overs.
By grounding candidate evaluations in verified high-leverage delivery data rather than unadjusted career totals, the Rajasthan Royals analytics group can systematically support informed squad construction, refine valuation models, and make evidence-backed auction decisions for IPL 2027.